

DATA-DRIVEN PRICING STRATEGY REPORT

For: Karmic Seed – Operations Analyst Round 3

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Tool Used: Microsoft Excel (Book1.xlsx), python.

1. EXECUTIVE SUMMARY

XYZ Company faced inconsistent pricing, margin erosion, and inventory imbalance due to reactive pricing decisions.

This project designs and implements a structured **data-driven pricing engine** that automatically recommends optimal prices for each SKU based on cost, margin constraints, competitor benchmarks, inventory health, demand strength, and advertising efficiency.

The model produces **final recommended prices that maximize margin while ensuring market competitiveness and inventory control.**

2. DATASETS USED

Dataset	Purpose
Pricing_Data	Cost & margin constraints
Competitor_Data	Market benchmark prices
Historical_Sales	Demand measurement
Inventory_Health	Stock pressure detection
Ads_Performance	Advertising efficiency

3. KEY BUSINESS SIGNALS IDENTIFIED

Signal	Business Meaning
Cost + Fees	Minimum sustainable price
Minimum Margin %	Loss prevention
Target Margin %	Profit objective
Competitor Price	Market alignment
Units Ordered	Demand strength
Sell-Through	Sales velocity

Signal	Business Meaning
Days of Supply	Inventory urgency
ROAS / ACOS	Advertising profitability

4. PRICING STRATEGY FRAMEWORK

Step-1: Cost Floor Protection

$\text{Min_Allowed_Price} = (\text{Cost} + \text{FBA} + \text{Storage} + \text{Handling}) / (1 - \text{Minimum_Margin}\%)$

Step-2: Profit Target Price

$\text{Target_Price} = (\text{Cost} + \text{Fees}) / (1 - \text{Target_Gross_Margin}\%)$

Step-3: Competitor Alignment

Condition	Adjustment
Competitor Price < Target	Slight reduction
Competitor Price > Target	Maintain / increase

Step-4: Inventory & Demand Logic

Scenario	Pricing Action
Low stock & High demand	Increase price
High stock & Low demand	Decrease price
Normal	Maintain

Step-5: Advertising Performance

Ads Result	Action
High ROAS	Increase price
High ACOS	Reduce price

5. IMPLEMENTATION TOOL

Platform: Microsoft Excel

Output: Book1.xlsx

Automated calculation of:

- Min Allowed Price
- Target Price
- Competitor Adjusted Price

- Inventory Adjusted Price
- Final Recommended Price

6. FINAL PRICING OUTPUT

SKU	Base_Cost	Min_Price	Target_Price	Avg_Competitor_Price	Total_Units_Ordered	Conversion_Rate	Days_of_Supply	Inventory_Status	ACOS	Returns_30D	High_Returns_Flag	Pricing_Decision	Recommended_Price
MN-25	16	31.27255	31.3196187	35.5	13121	4471	1645	341	0.203699	73	High Returns	Decrease	33.725
MN-26	13	26.90381	26.9443051	30.4	2961	1663	1142	1079	0.169361	30	Normal	Keep	30.4
MN-27	11	24.22846	24.2649272	29.9	2609	1257	838	772	0.20589	21	Normal	Keep	29.9
MN-28	11	22.51503	22.5489212	26.2	5516	2480	1327	1043	0.228958	45	High Returns	Decrease	24.89
MN-29	8	17.74549	17.7722027	22.8	5346	1519	924	815	0.230921	72	High Returns	Decrease	21.66
MN-30	16	35.3006	35.3537381	36.4	3771	688	404	254	0.261357	8	Normal	Keep	36.4
MN-31	13	26.60321	26.6432514	29.8	1584	355	300	199	0.276398	7	Normal	Keep	29.8
MN-32	11	23.95792	23.9939789	27.5	1484	533	338	230	0.169492	16	Normal	Keep	27.5
MN-33	11	22.20441	22.2378324	0	1184	388	326	153	0.163605	13	Normal	Keep	22.20440882
MN-34	14	28.49699	28.5398896	33.8	2106	1001	753	498	0.194233	14	Normal	Keep	33.8
MN-35	14	28.54709	28.5900652	32.9	1418	417	267	168	0.208198	8	Normal	Keep	32.9
MN-36	11	25.1002	25.1379829	30.1	11890	2255	1382	784	0.290455	36	High Returns	Decrease	28.595
MN-37	13	25.97194	26.0110386	29.4	1160	252	221	123	0.311108	9	Normal	Keep	29.4
MN-39	13	22.97595	23.0105369	25.8	1176	282	0	22	0.315135	4	Normal	Keep	25.8
MN-40	9	19.70942	19.7390868	24.9	2333	883	655	407	0.124316	38	High Returns	Decrease	23.655
MN-41	41	63.54709	63.6427496	82.5	2779	1613	935	694	0.087023	12	Normal	Keep	82.5
MN-42	31	50.89178	50.9683894	76.8	2608	1267	752	673	0.120939	28	Normal	Keep	76.8
MN-43	4	10.83166	10.8479679	14.8	1707	207	166	123	0.61236	5	Normal	Keep	14.8
MN-44	3	9.679359	9.69392875	0	1139	135	130	96	0.262046	7	Normal	Keep	9.679358717
MN-45	14	25.73146	25.7701957	31.4	1046	358	159	82	0.084085	3	Normal	Keep	31.4
MN-46	11	21.03206	21.063723	28.9	736	173	177	109	0.232348	3	Normal	Keep	28.9
MN-47	34	55.4509	55.5343703	80.4	834	306	256	241	0.106789	10	Normal	Keep	80.4

SKU Cost Target Price Competitor Price Final Recommended Price

7. KEY BUSINESS BENEFITS

Benefit	Impact
Margin Protection	Prevents loss
Inventory Balance	Avoids stockouts & overstock
Market Competitiveness	Aligns with competition
Automation Ready	Scales to 1000+ SKUs

8. CONCLUSION

This pricing engine converts raw operational data into structured pricing actions. It enables XYZ Company to move from reactive pricing to **predictable, profitable, and scalable pricing management**.

9. ATTACHED FILES

- Sahil_Bhoir_Pricing_Model.xlsx
- Pricing Calculation Sheet
- karmic_seed_pricing_analysis.ipynb